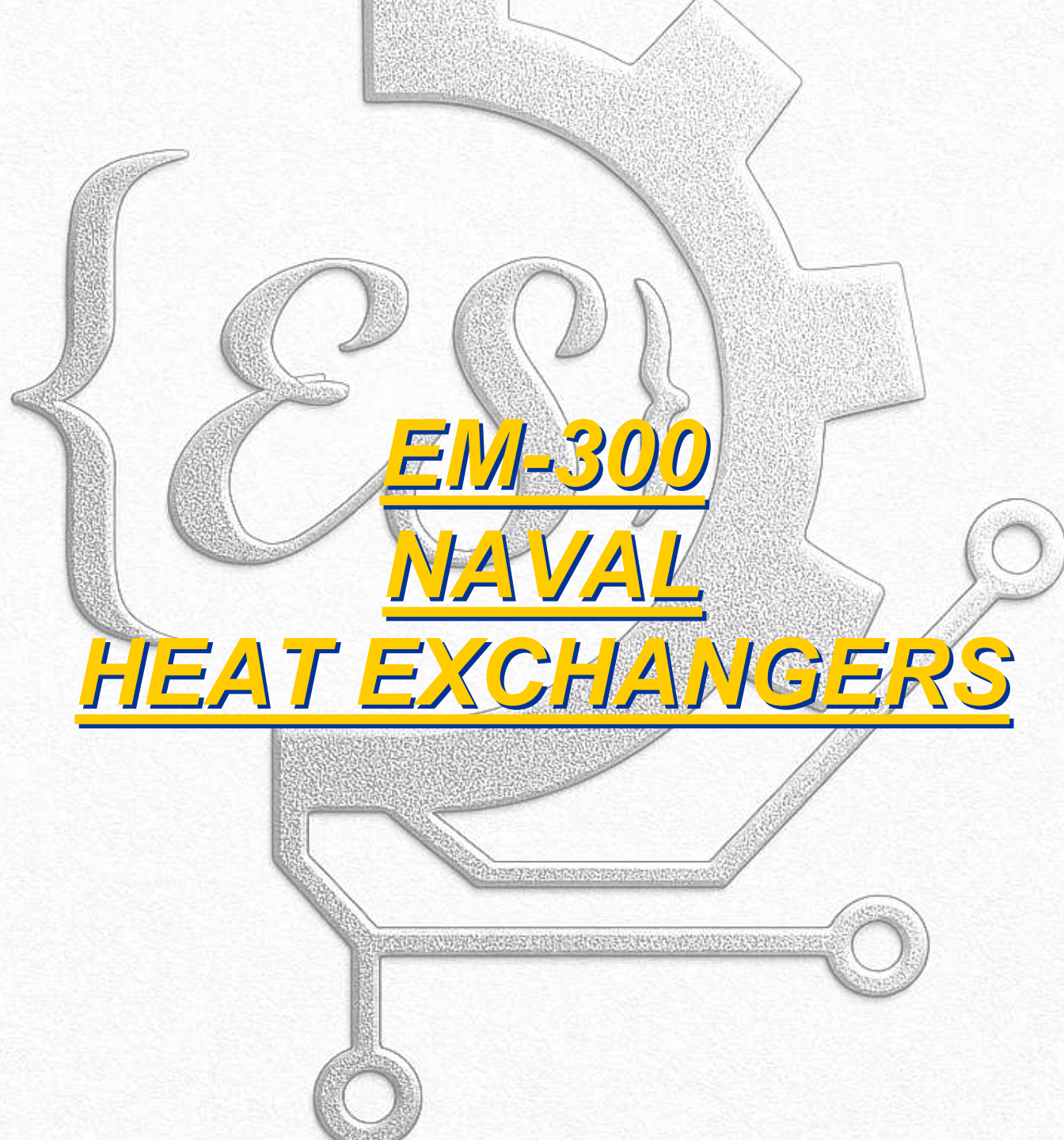


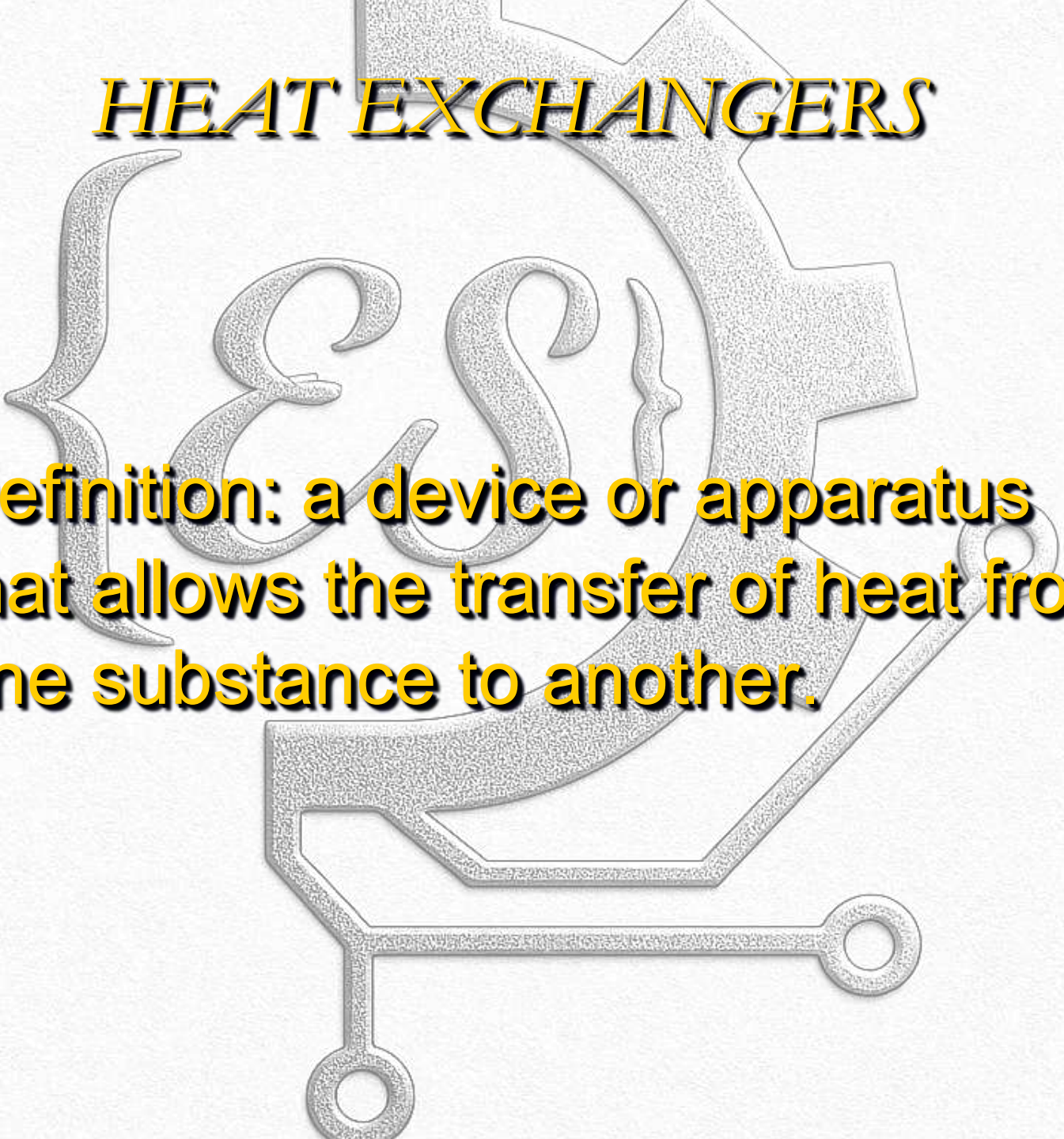


EM-300
NAVAL
HEAT EXCHANGERS

ENABLING OBJECTIVES

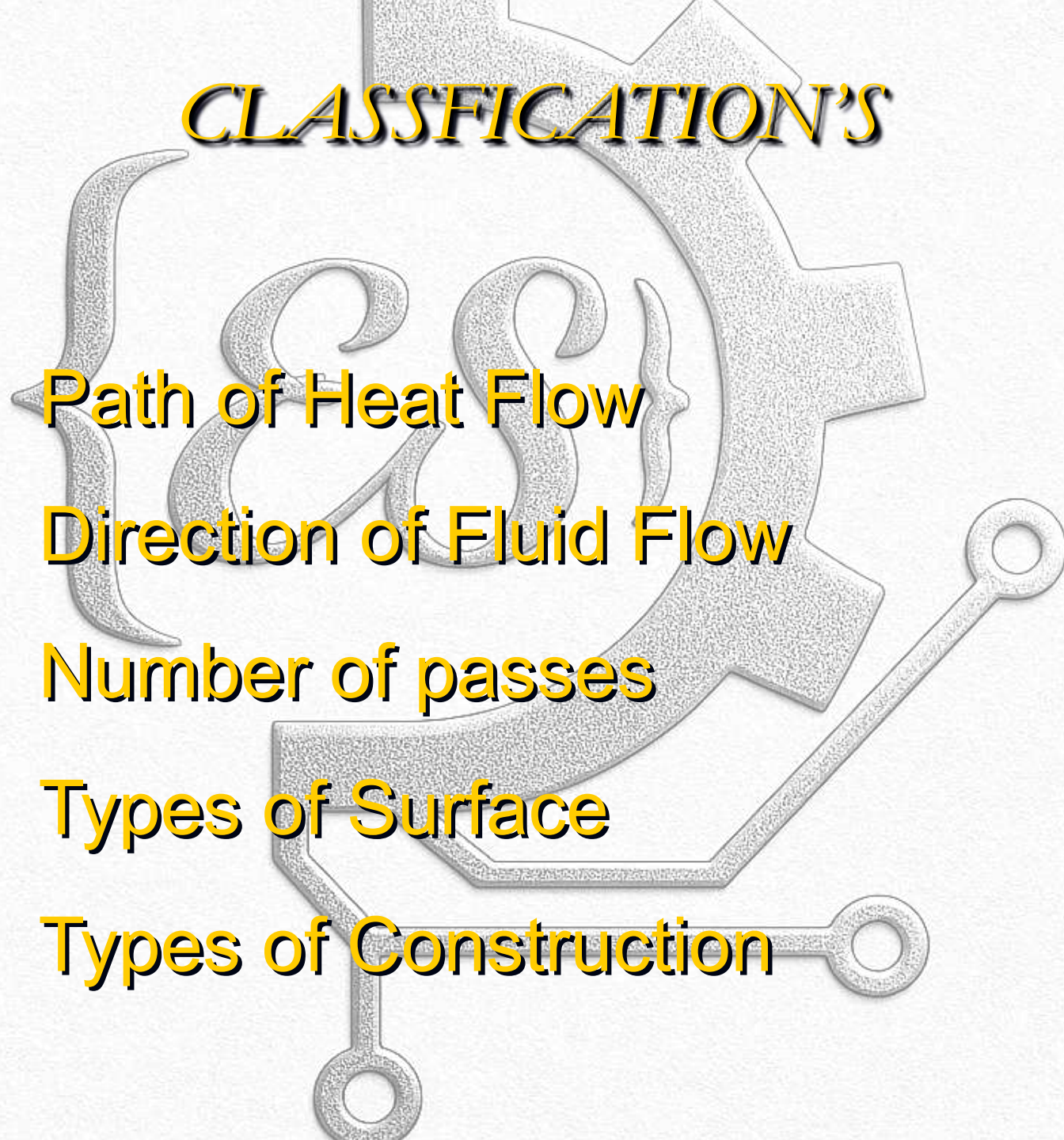
- **Understand the basic operation of a heat exchanger in accordance with Machinist Mate 3 & 2, NAVEDTRA 14151.**
- **Understand the different classification methods of heat exchangers including; Path of heat flow, Direction of Fluid flow, Number of passes, Types of Surface, and Types of Construction in accordance with Machinist Mate 3 & 2, NAVEDTRA 14151.**
- **Understand the most common naval uses of heat exchangers in accordance with Machinist Mate 3 & 2, NAVEDTRA 14151.**

HEAT EXCHANGERS

A large, stylized, metallic-looking graphic in the background. It features a large gear-like shape at the top, a large curly brace on the left, and a complex, circuit-like or pipe-like structure at the bottom. The entire graphic has a textured, metallic appearance.

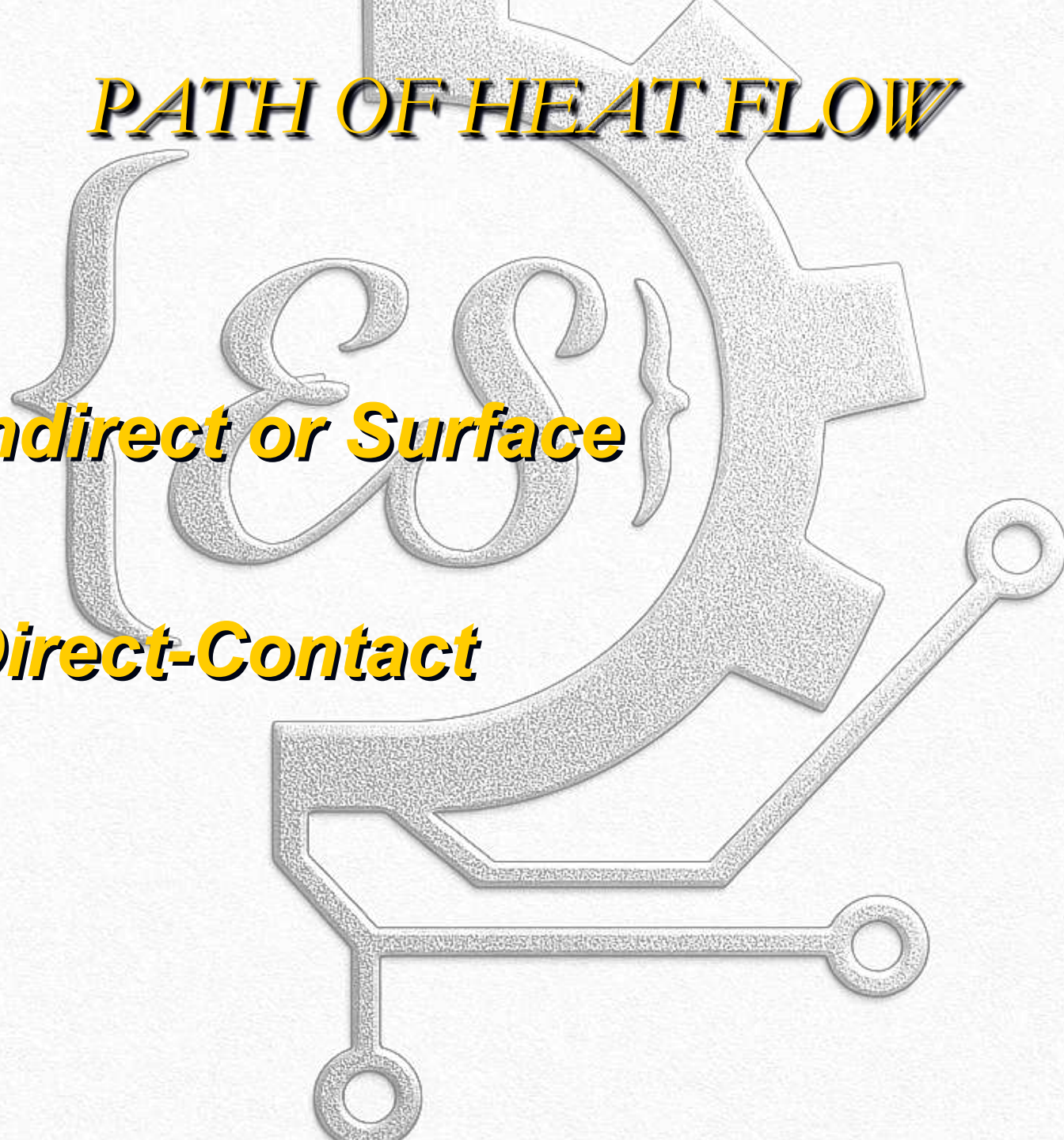
Definition: a device or apparatus that allows the transfer of heat from one substance to another.

CLASSIFICATION'S



- Path of Heat Flow
- Direction of Fluid Flow
- Number of passes
- Types of Surface
- Types of Construction

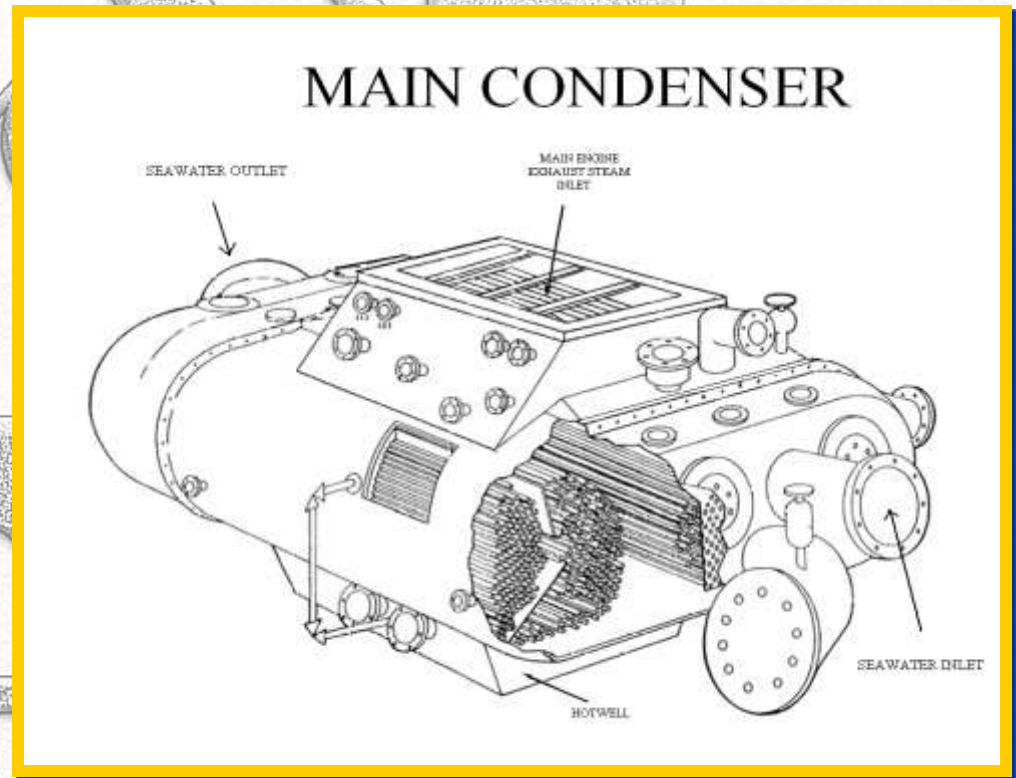
PATH OF HEAT FLOW



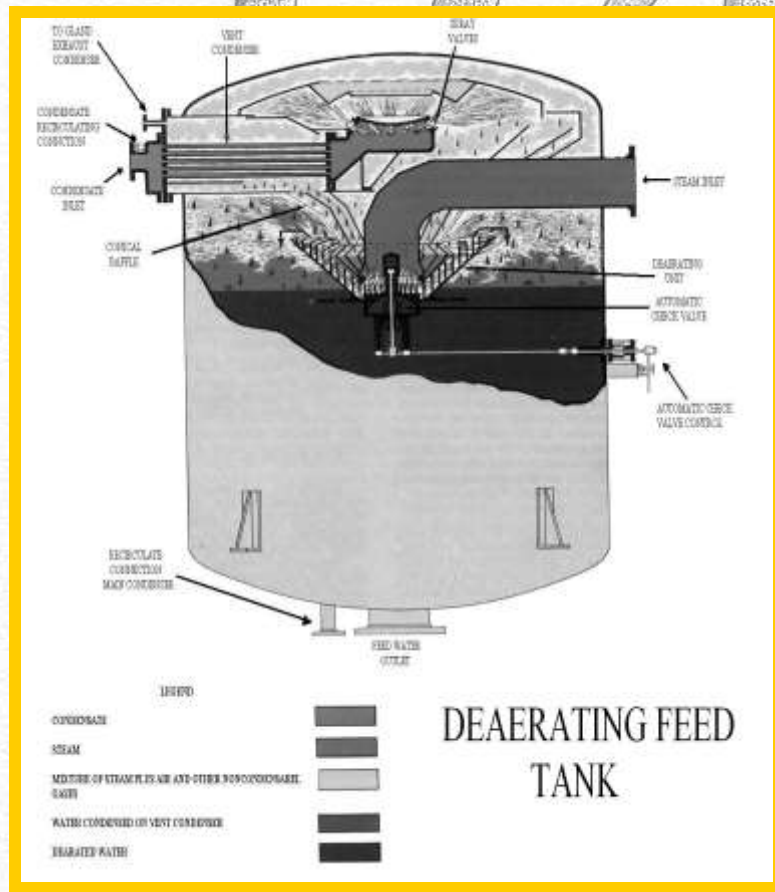
- ***Indirect or Surface***
- ***Direct-Contact***

INDIRECT CONTACT TYPE

Heat flows from one fluid to the other through a tube, plate, or other surface that separates the two fluids.



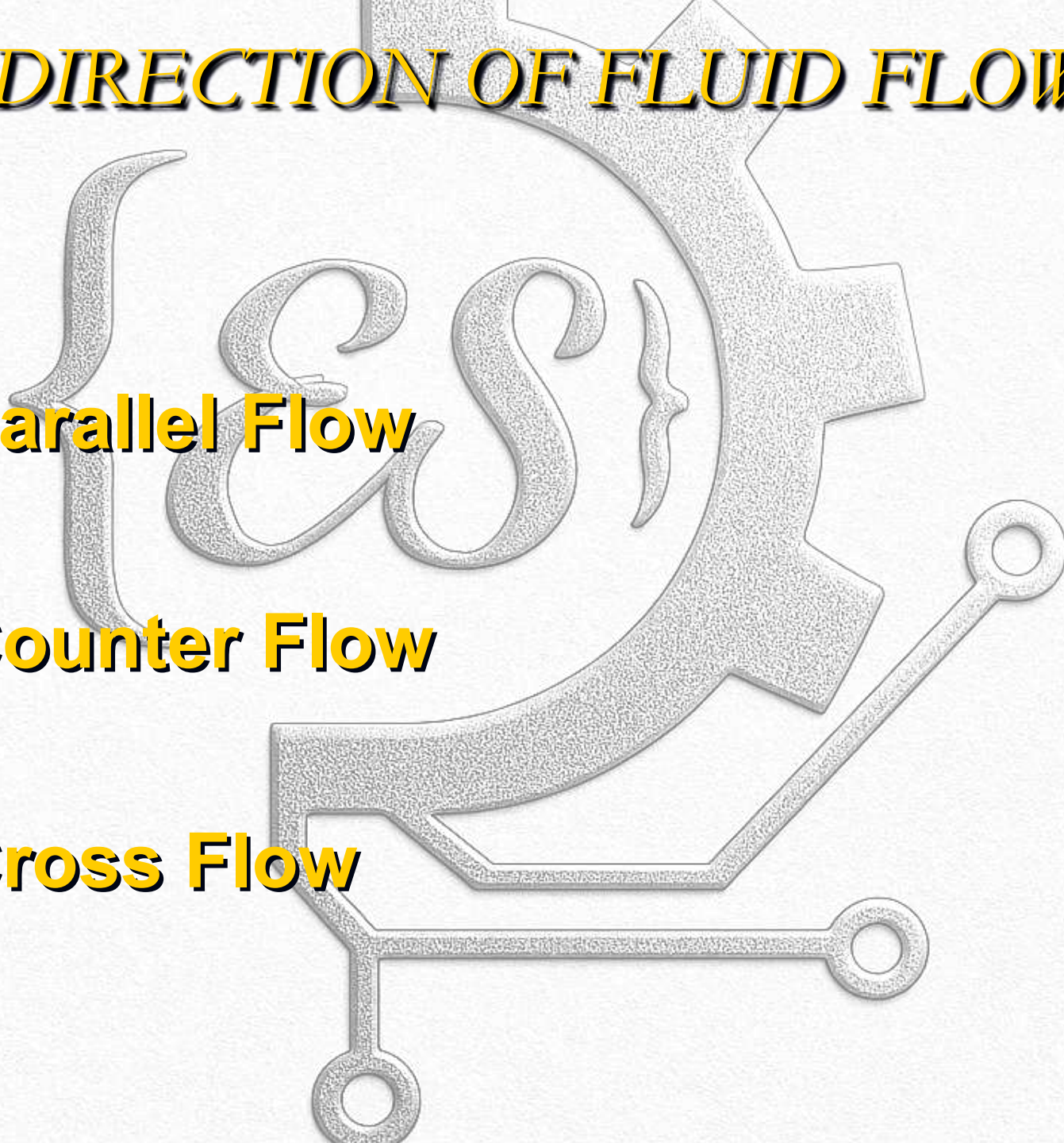
DIRECT-CONTACT TYPE



- Heat is transferred directly from one fluid to another as the two fluids mix.

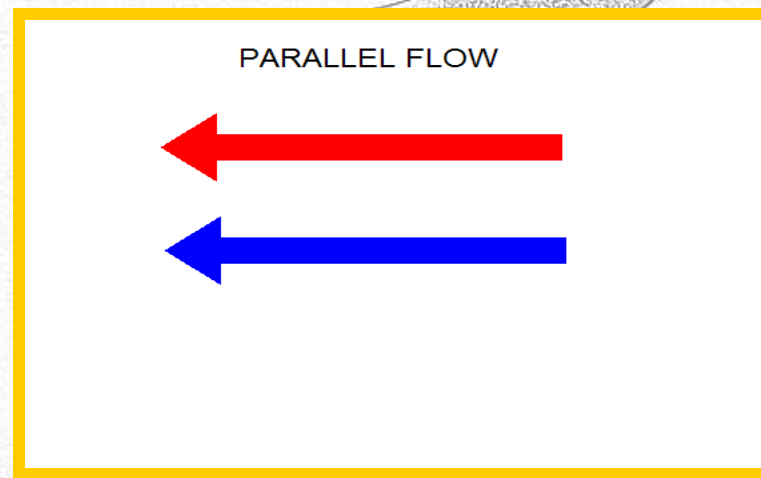
DIRECTION OF FLUID FLOW

- **Parallel Flow**
- **Counter Flow**
- **Cross Flow**



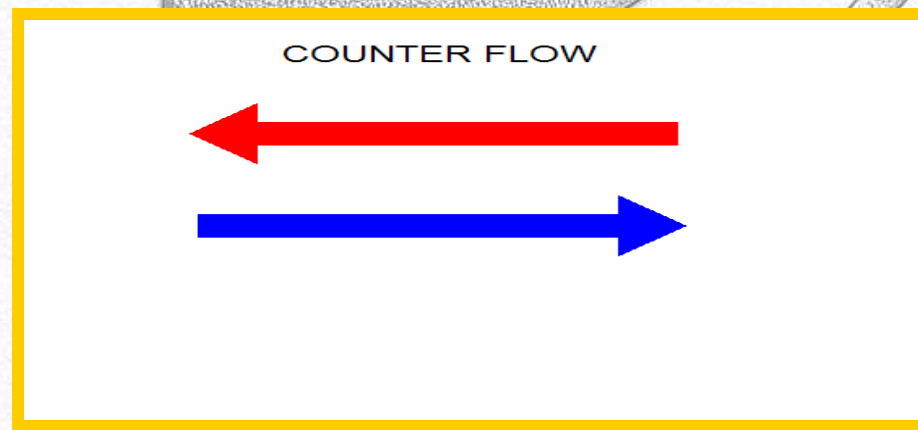
PARALLEL FLOW

- Fluids flow in the same direction.
- Large temperature difference at inlet and a small temperature difference at outlet.
- Higher heat transfer rate at the inlet than the outlet.

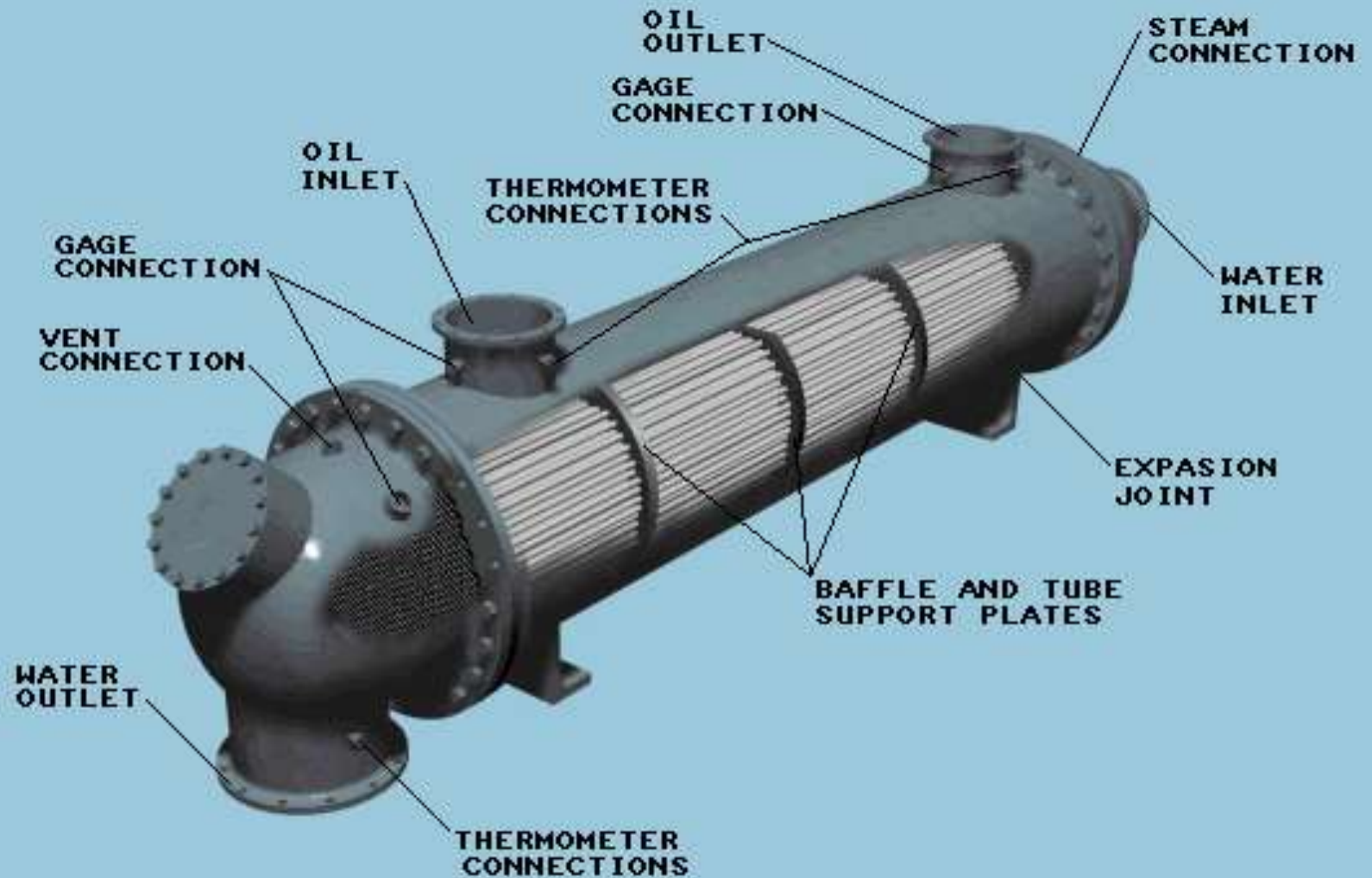


COUNTER FLOW

- Two fluids flow in opposite directions.
- Less area is required than parallel flow because there is a large enough temperature difference maintained from the inlet to the outlet.

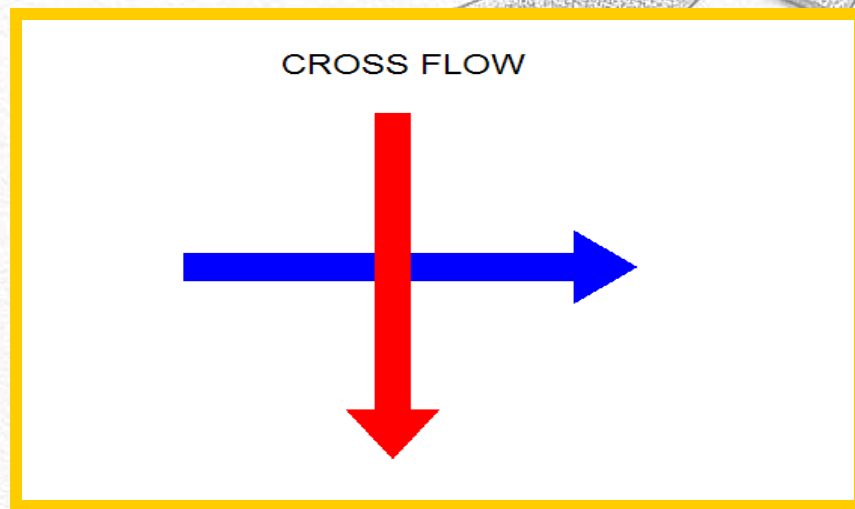


Main Lube Oil Cooler



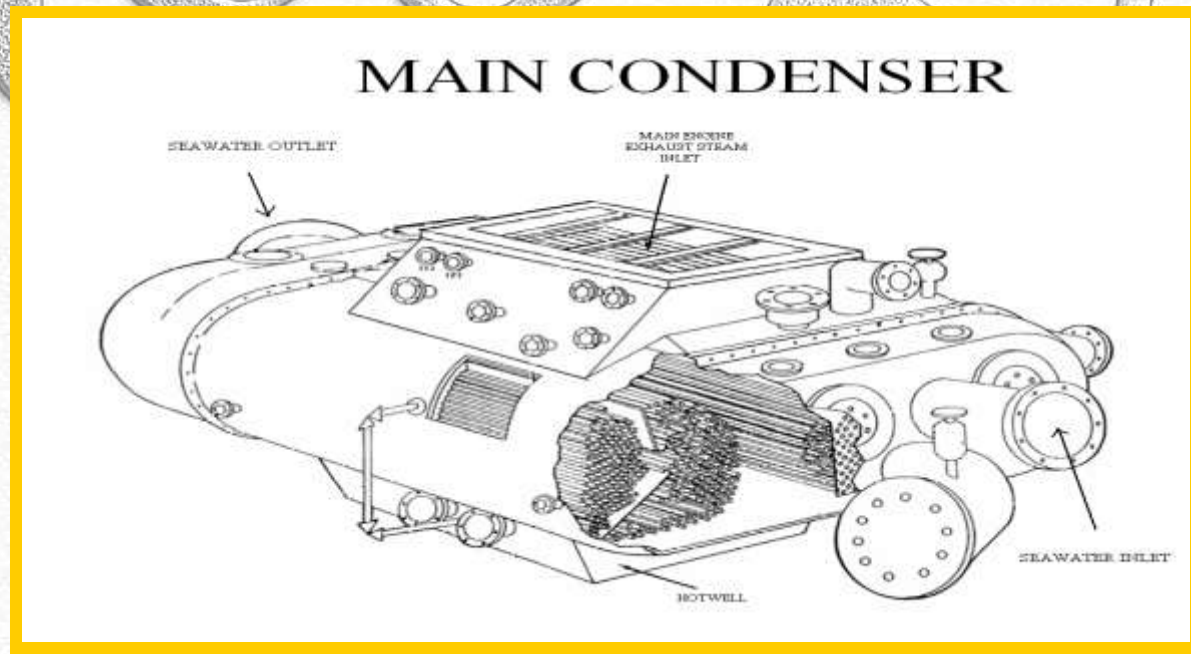
CROSS FLOW

- **Fluid flows at right angles of each other.**
- **Cross flow is used for most condensers because it is effective in removing latent heat.**



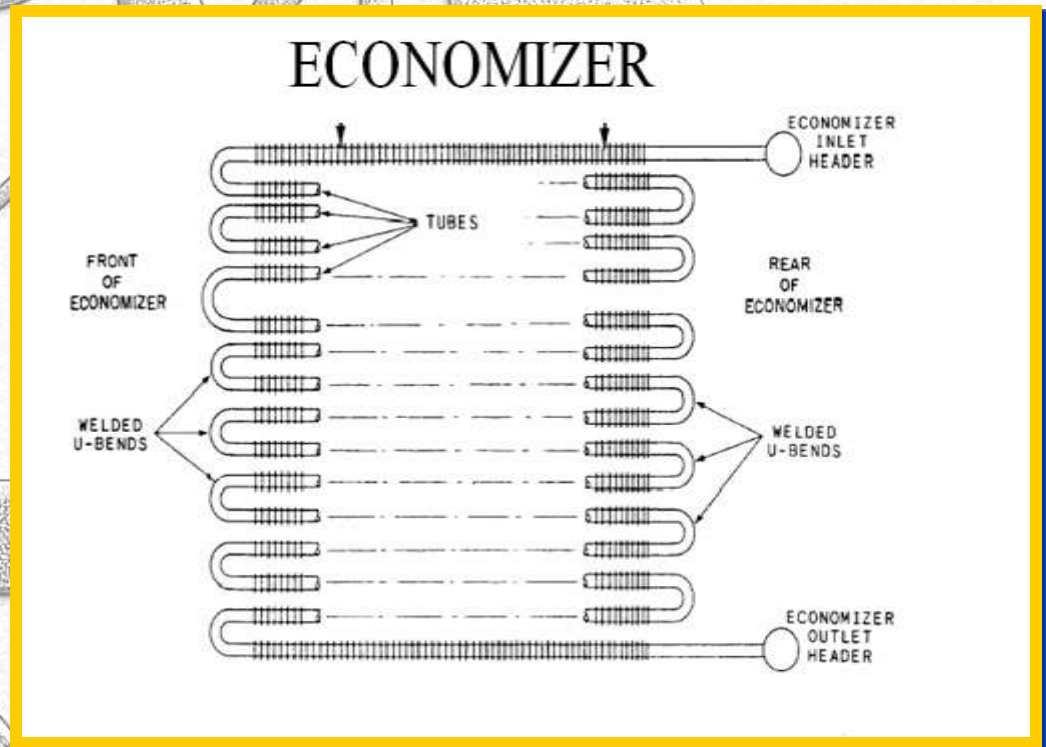
NUMBER OF PASSES

- **Single Pass:** The fluids pass each other only once.

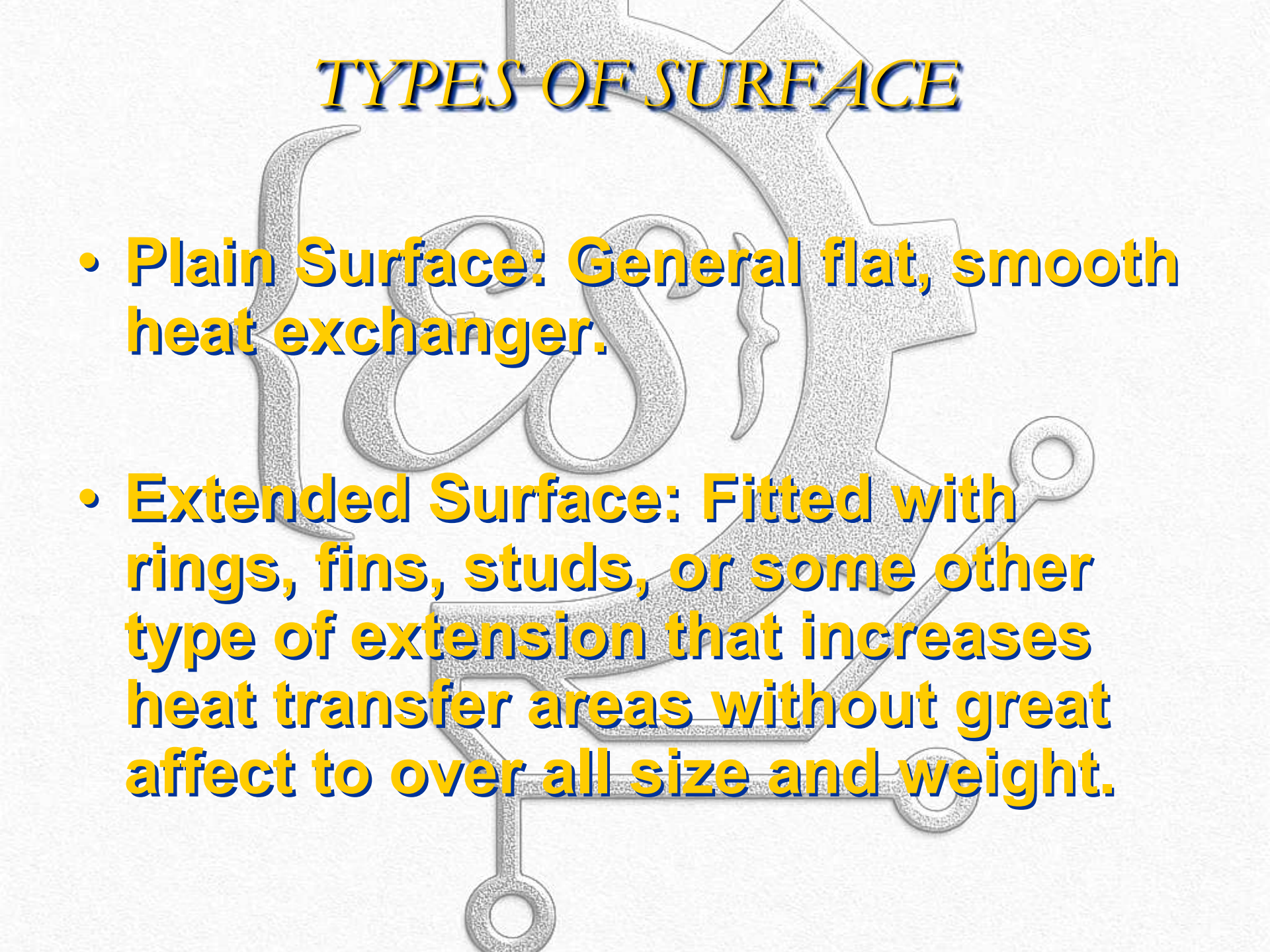


NUMBER OF PASSES

- **Multipass:** The fluids pass each other more than once.

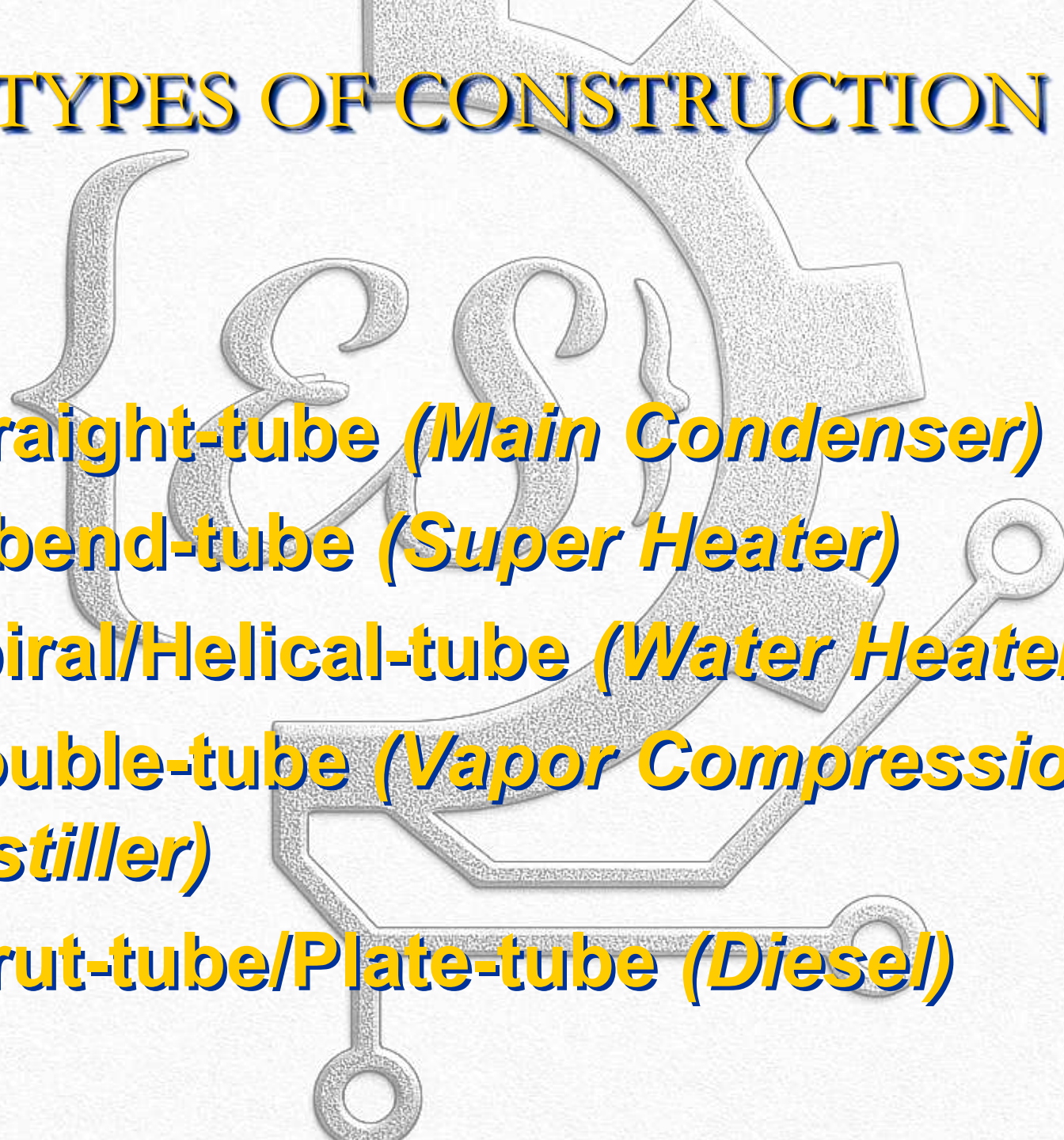


TYPES OF SURFACE

A faint, stylized background graphic featuring a large gear and a wrench, suggesting a mechanical or engineering theme.

- **Plain Surface:** General flat, smooth heat exchanger.
- **Extended Surface:** Fitted with rings, fins, studs, or some other type of extension that increases heat transfer areas without great affect to over all size and weight.

TYPES OF CONSTRUCTION



- **Straight-tube** (*Main Condenser*)
- **U-bend-tube** (*Super Heater*)
- **Spiral/Helical-tube** (*Water Heater*)
- **Double-tube** (*Vapor Compression Distiller*)
- **Strut-tube/Plate-tube** (*Diesel*)

NAVAL APPLICATIONS

The background of the slide features a large, light gray gear with a ship's anchor superimposed over it. The gear is positioned in the upper half, and the anchor is in the lower half, both rendered in a metallic, textured style.

- Surface Ships (*Lube Oil Coolers/Radar Systems*)
- Aviation (*Electronic Cooling Systems/Fuel-Oil Heat Exchangers*)
- Tanks (*Oil Coolers*)
- Submarines (*Lube Oil Coolers/ Electronic and Sonar cooling Systems*)